

STATISTICAL INVESTIGATION OF CHRONIC KIDNEY DISEASE RISK FACTORS AND PREDICTION

Complex Engineering
Project (CEP) —
STA281 Spring 2026

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01 — Objective: Identify CKD Risk Factors & Predictive Insights

02 — Dataset: 400 Patients, 26 Variables (Kaggle)

03 — Methodology: Descriptive Stats, Probability, Hypothesis Testing, Regression

04 — Target: Binary CKD Classification



CKD

STA281 CEP

03 - DATASET OVERVIEW

400 patient records across 26
clinical and laboratory variables

Target Variable: CKD status (binary: 1 = CKD, 0 = No CKD)

Variables include clinical measurements such as serum creatinine, blood urea, hemoglobin, and blood pressure, alongside demographic and lab indicators.

Preprocessing Steps: Variable renaming for clarity, missing value imputation, numeric type conversions, and target label standardization to prepare data for statistical analysis.

400 Records

26 Variables

Key Clinical Variables — CKD Patient Dataset (n=400)

Serum Creatinine: Mean 3.07 | Median 1.80 | Std Dev 3.31 | Range: 0.5–15.2

Blood Urea: Mean 57.4 | Median 44.0 | Std Dev 40.6 | Range: 10–391

Hemoglobin: Mean 12.5 | Median 13.9 | Std Dev 2.91 | Range: 3.1–17.8

Blood Pressure: Mean 76.5 | Median 80.0 | Std Dev 13.9 | Range: 50–180



Elevated creatinine & urea indicate impaired filtration; low hemoglobin signals anemia risk in CKD patients.

04 — DESCRIPTIVE STATISTICS SUMMARY

05 - VISUALIZATION FINDINGS

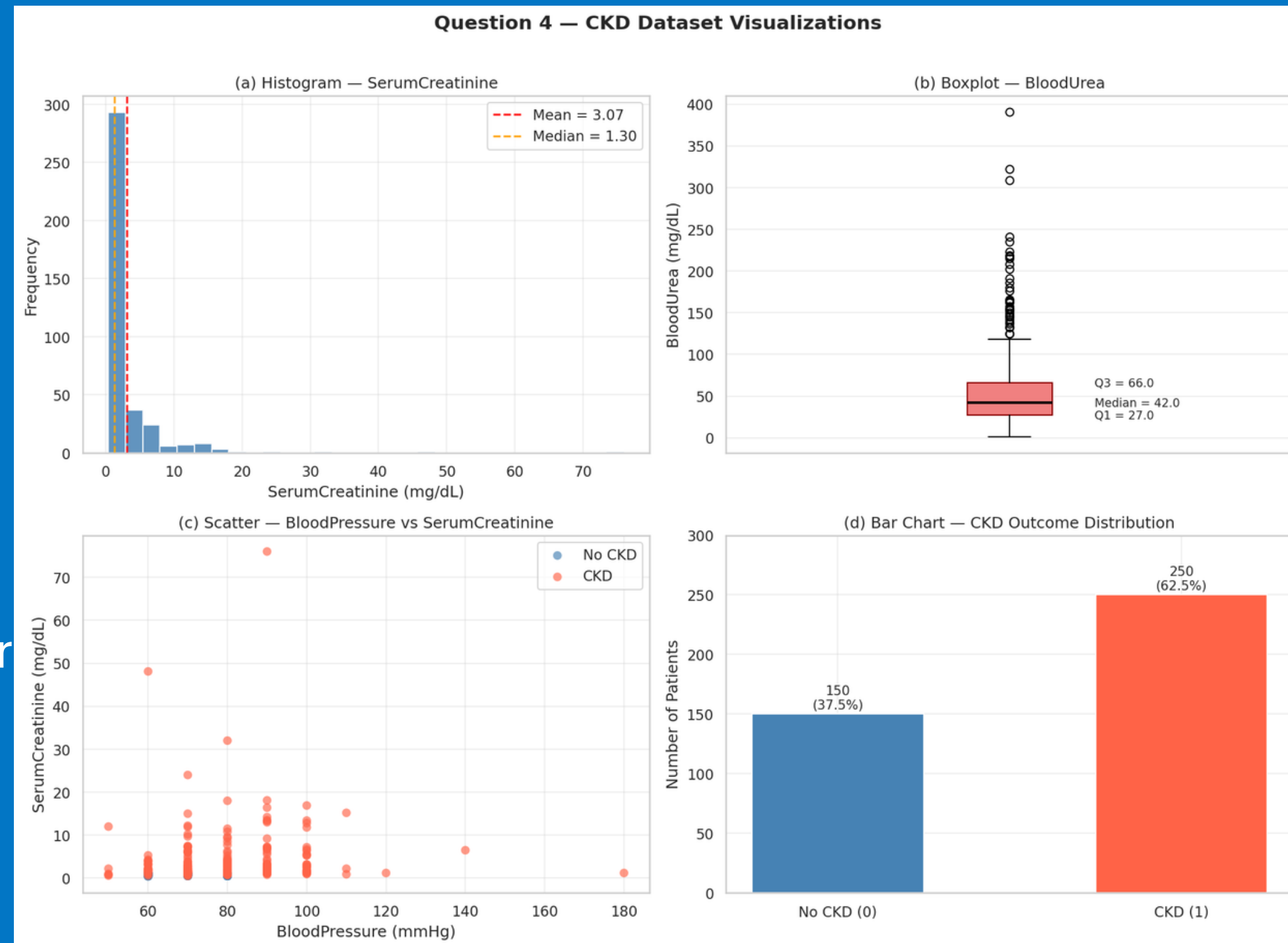
CKD prevalence: 62.5% vs 37.5% non-CKD in the dataset

(a) SerumCreatinine histogram shows right-skew — some patients with severe kidney dysfunction.

(b) BloodUrea boxplot reveals wide spread and outliers indicating abnormal urea accumulation.

(c) BP vs Creatinine scatter: CKD patients cluster at higher values.

(d) CKD outcome bar chart confirms 62.5% CKD prevalence.



06 - CKD VS NON-CKD COMPARISON

Key differences between CKD and Non-CKD groups:

- SerumCreatinine: Significantly higher in CKD patients
- BloodUrea: Elevated, indicating impaired filtration
- BloodPressure: Higher in CKD — hypertension link
- Hemoglobin: Lower in CKD — anemia indicator

**CKD PATIENTS SHOW
ELEVATED KIDNEY
MARKERS AND LOWER
HEMOGLOBIN**

Clinical differentiation supports early CKD screening and diagnosis

CKD

Non-CKD

07 - PROBABILITY & BAYES ANALYSIS

Conditional Probability:
 $P(\text{CKD} \mid \text{SerumCreatinine} > 1.5) = 100\%$

Bayes Theorem Result:
Posterior probability = 100% for threshold $> 1.8 \text{ mg/dL}$

Clinical Implication:
Elevated SerumCreatinine is a near-perfect predictor of CKD status, confirming its role as a primary screening biomarker.

Bayes probability



Elevated SerumCreatinine strongly predicts CKD with 100% posterior probability

Bayes

$P(\text{CKD}) = 100\%$

08 - PROBABILITY DISTRIBUTIONS & CLT

Five distributions model CKD data patterns:

- Binomial PMF: CKD counts in sample ($n=12$, $p=0.55$)
- Normal: SerumCreatinine risk categories
- Uniform: Patient age probabilities
- Exponential: CKD admission intervals
- CLT: Sampling distribution of BloodUrea mean converges to normality

Binomial & Normal distributions capture CKD prevalence and biomarker spread across patient populations

Binomial

CLT Applied

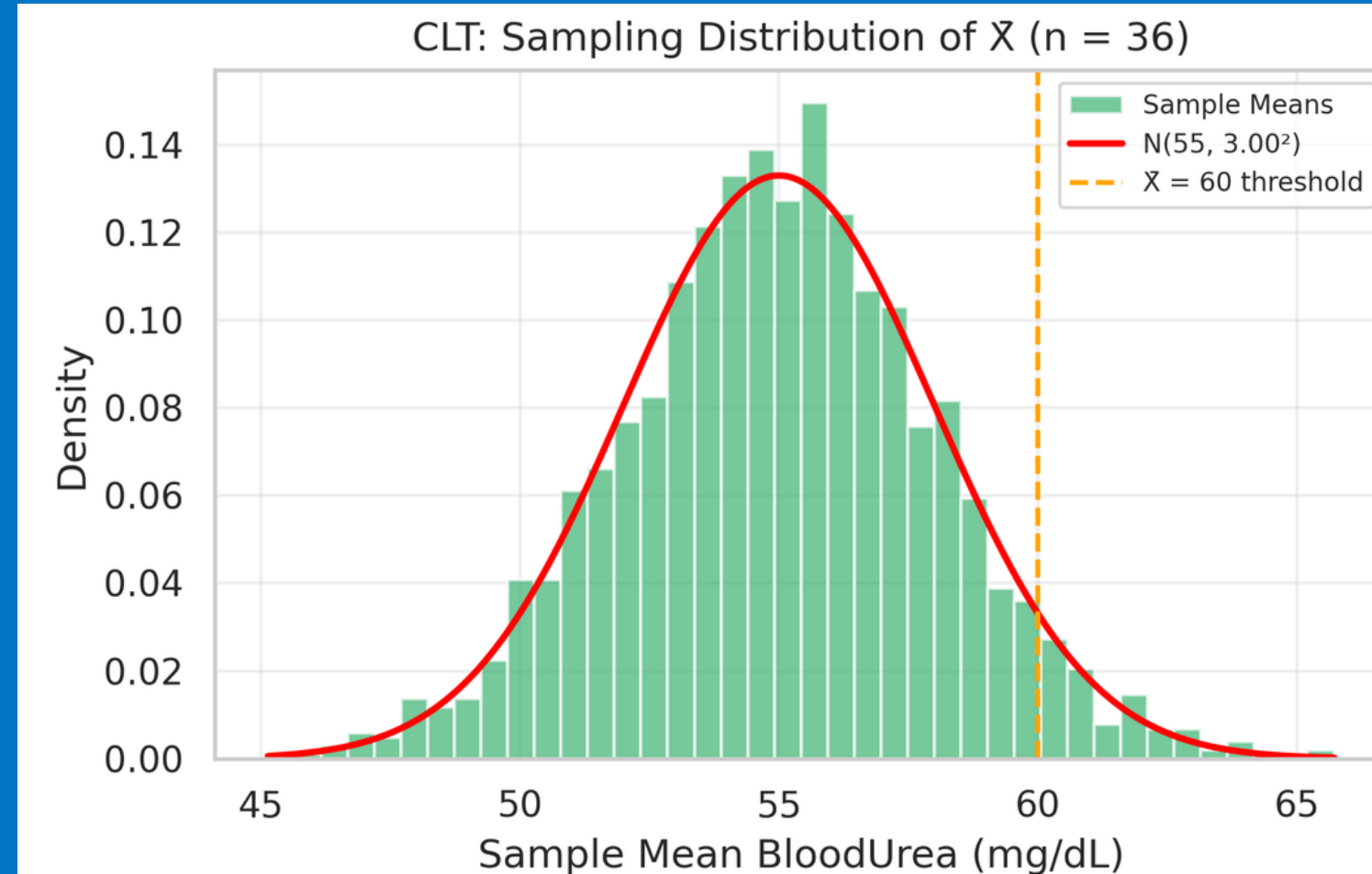
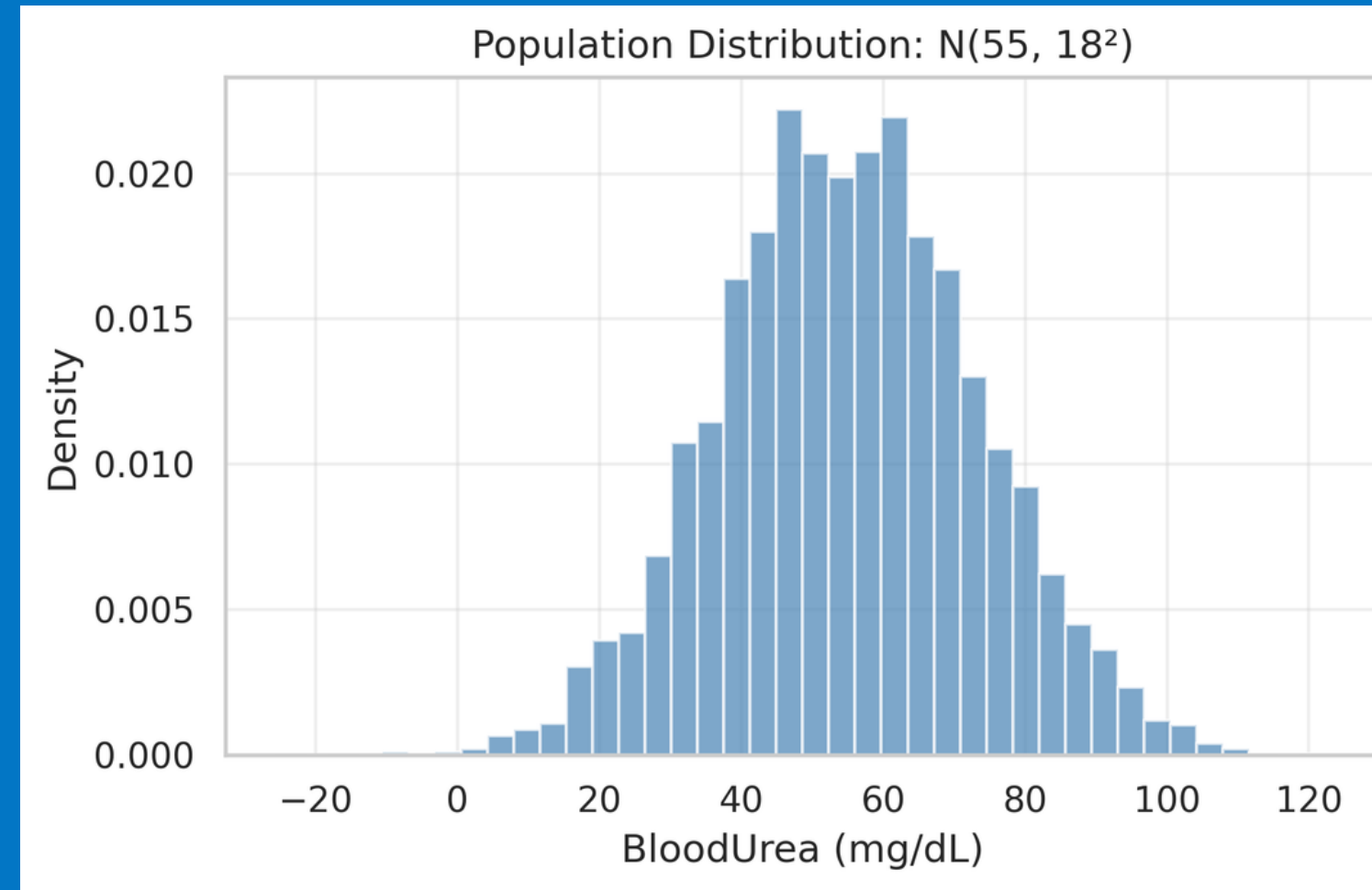
CLT confirms BloodUrea sample means follow normal distribution, enabling robust statistical inference

09 - DISTRIBUTIONS & CLT

Exponential distribution models CKD admission intervals; CLT confirms sample mean convergence for robust inference

Exponential Distribution: Models time between CKD patient admissions, revealing clustering patterns in high-risk periods.

Central Limit Theorem: Sampling distribution of BloodUrea mean converges to normal as sample size increases, validating parametric inference methods used throughout this study.



10 - CONFIDENCE INTERVAL & HYPOTHESIS TESTING

95% CI for SerumCreatinine: Mean above normal range — indicates impaired kidney filtration.

Z-Test (Right-tailed): BloodUrea significantly > 50 mg/dL — confirms abnormal urea accumulation.

T-Test (Left-tailed): Hemoglobin significantly < 12.5 g/dL — confirms anemia in CKD patients.

Statistical evidence confirms impaired kidney function in CKD patients

All three tests support data-driven diagnosis of chronic kidney dysfunction

CI

Hypothesis

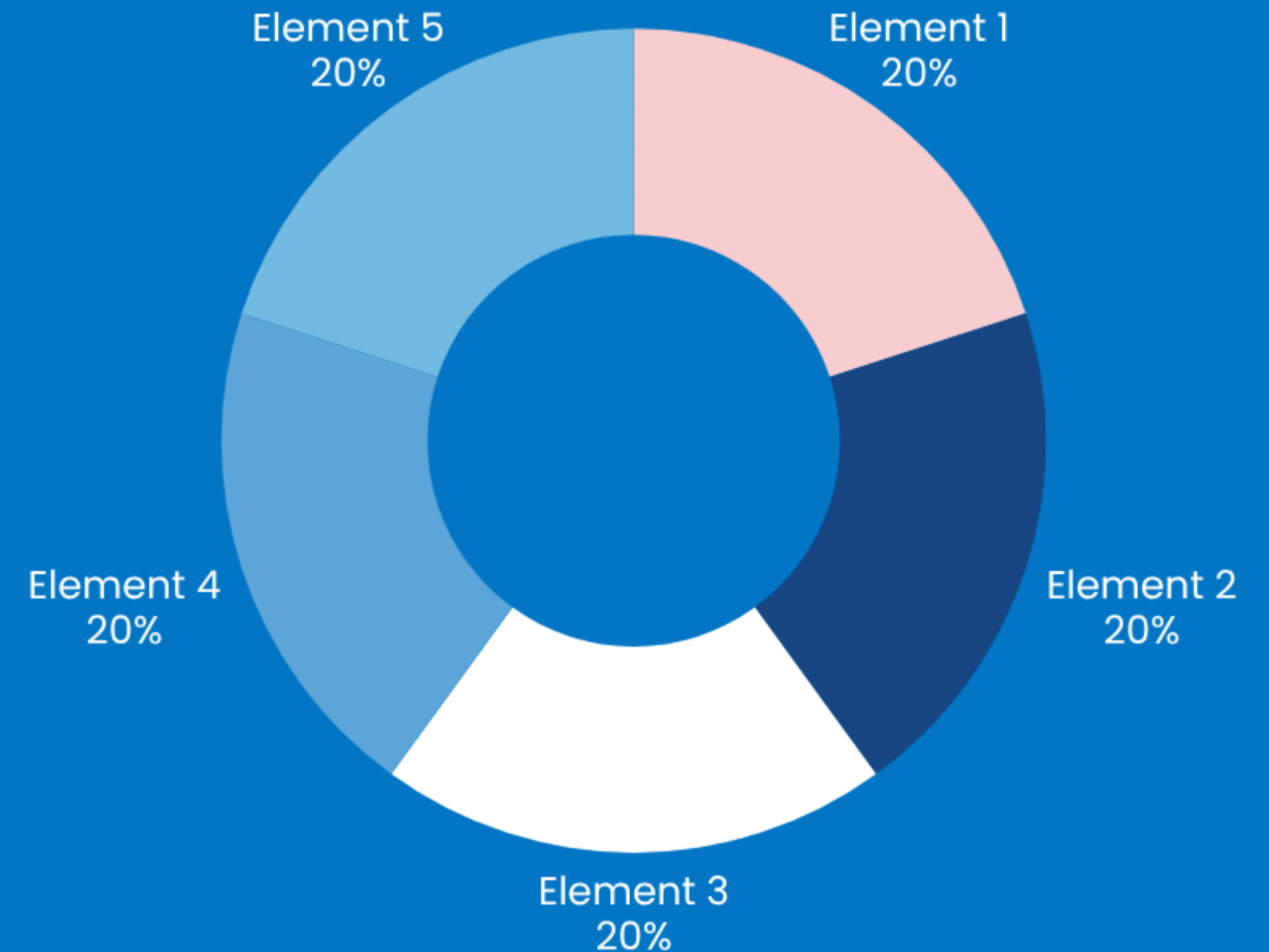
10 - CORRELATION ANALYSIS

Key Correlations:

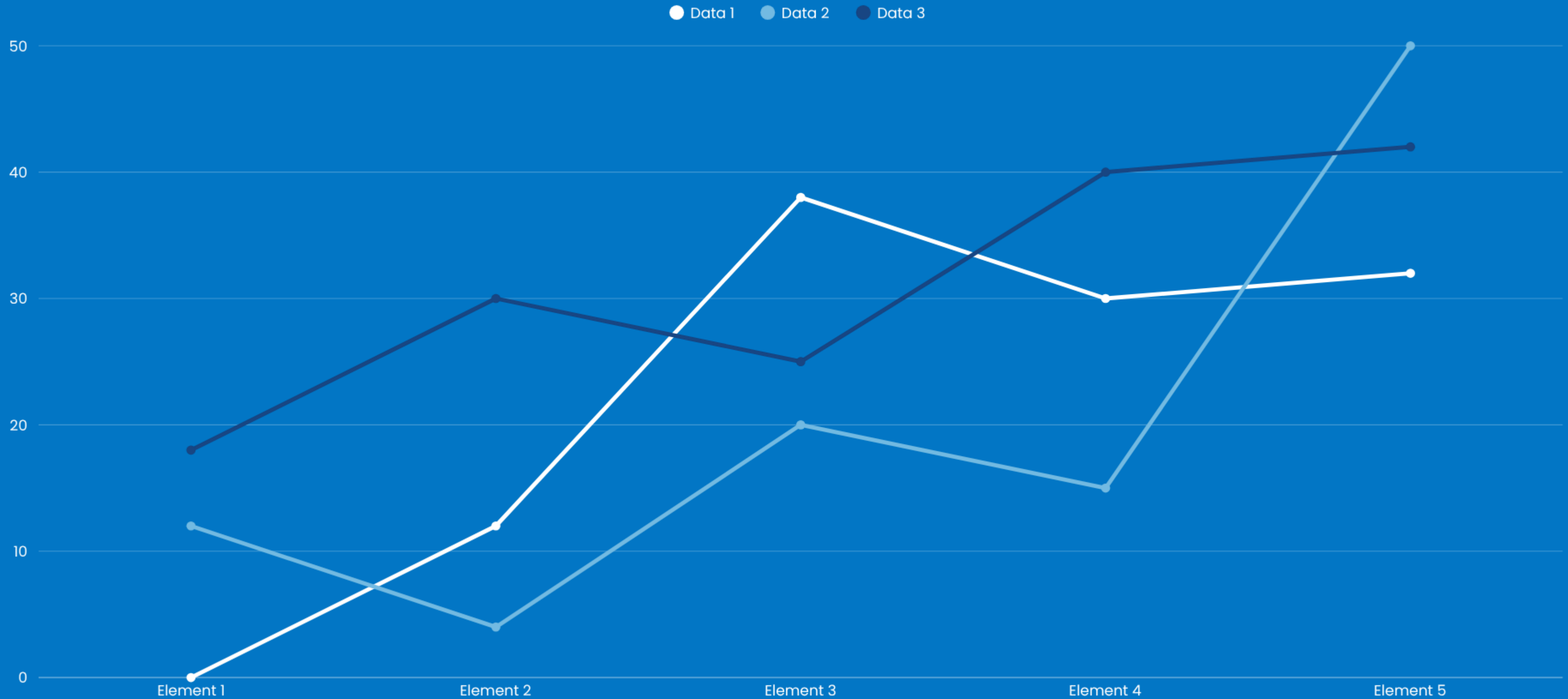
- SerumCreatinine & BloodUrea: $r = +0.78$ (Strong Positive)
Elevated creatinine and urea rise together, confirming impaired kidney filtration.
- Hemoglobin & CKD: $r = -0.81$ (Strong Negative)
Lower hemoglobin strongly associated with CKD presence, indicating anemia.
- BloodPressure & CKD: Moderate positive correlation
Hypertension linked to progressive kidney damage.

Clinical Insight:

Correlation analysis confirms expected clinical relationships between kidney function markers and CKD status, supporting their use as diagnostic indicators.



11 - Regression & Predictive Insights



01 — SerumCreatinine & BloodUrea

02 — Hemoglobin &
BloodPressure

03 — Hypertension, Diabetes, Electrolyte
Imbalance

04 — Elevated Creatinine & Low Hemoglobin

Biomarkers

Risk Factors



- CKD patients show significantly altered clinical markers across all key biomarkers
- Probability & Bayes analysis confirm SerumCreatinine as a near-perfect CKD predictor
- Distribution modeling & CLT support robust statistical inference on patient data
- Hypothesis testing confirms abnormal BloodUrea and Hemoglobin in CKD patients
- Correlation analysis reveals strong links between kidney markers and CKD status
- Comprehensive statistical analysis supports reliable, data-driven CKD prediction



Statistical evidence strongly supports data-driven prediction of CKD risk factors.

14 - EVIDENCE-BASED CONCLUSION

15 - RECOMMENDATIONS



- ✓ Regularly monitor SerumCreatinine, BloodUrea, BloodPressure, and Hemoglobin in at-risk patients.
- ✓ Initiate early screening for hypertensive and diabetic patients before symptom onset.
- ✓ Control blood pressure and manage diabetes to reduce CKD progression risk.
 - ✓ Monitor anemia indicators and electrolyte levels for timely early intervention.
- ✓ Promote lifestyle modifications including diet, hydration, and exercise.
- ✓ Schedule regular follow-up testing to track biomarker trends over time.

Early detection and evidence-based management can slow CKD progression

Screen

Prevention

THANK YOU

